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Unit 1 : Session 5 : SLO 2 | Matrix Multiplication

I. Fill in the Blanks

- columns are valid only if columns of the first matrix equal rows of the second.
- rows of first matrix the result has dimensions (rows of first) $\times$ (columns of second).
- corresponding elements multiply each row element with the corresponding column element.
- each element (repeated/dot-product) a loop is required for each computation.
- nested loops use three nested loops (i, j, k).
- zero initialize result elements to 0.
- dot product the innermost loop calculates the dot product.
- accumulated (running sum) $\text{sum} += a[i][k] * b[k][j]$.
- result store the computed value in the result matrix $\text{res}[i][j]$.
- dimension mismatch (runtime/logical) error.
- 2D arrays (contiguous memory) matrix elements are stored in two-dimensional arrays.
- use (computation) initialize arrays before computing.
- dimensions declare the result with proper r1 $\times$ c2 dimensions.
- 2D array (2D) $\times$ (3D) $\rightarrow$ 2D.
- integer (element-wise multiply-accumulate) arithmetic.